PHASE 1 · MACHINE LEARNING · ANONYMOUS SUBMISSION

Leak-safe probabilistic demand forecasting on M5

Kalman filtering, LightGBM, a Kalman–LightGBM hybrid and DeepAR, compared under one rolling-origin protocol on 1,200 Walmart series.

Research question

Under a single leak-safe protocol, how do a local state-space model, a tabular gradient-boosting model, their hybrid and a global deep model compare on **accuracy**, **calibration** and **cost** for sparse retail demand?

Data

Frozen M5 subset · 3 stores · 1,000 days

Protocol

4 rolling origins · 28-day horizon · 7 quantiles

Scope

Phase 1 ML evaluation. No tuning was run, and each model uses one seed.

MOTIVATION

Ordering stock is a bet on a range of demand, not on a single number

Inventory needs quantiles

The newsvendor order level is a demand quantile:

$$q^* = C_u / (C_u + C_o)$$

C_u = cost of a lost sale, C_o = cost of holding a unit. If a lost sale costs 9× holding, q* = 0.9. We therefore score 7 quantiles, not only a mean.

Demand is sparse

59.11%

of launched, available cells have zero sales

90%

of series are intermittent or lumpy (1,081 of 1,200)

Some zeros are censored

A stockout records zero sales even when there was demand. M5 has **no inventory records**, so censoring can only be inferred with a heuristic rule.

Treating every zero as true demand teaches models to under-forecast.

The task: produce calibrated quantiles for sparse, partly censored counts without letting any model see the future.

EDA numbers: outputs/eda/dataset_summary.csv, intermittency_counts.csv

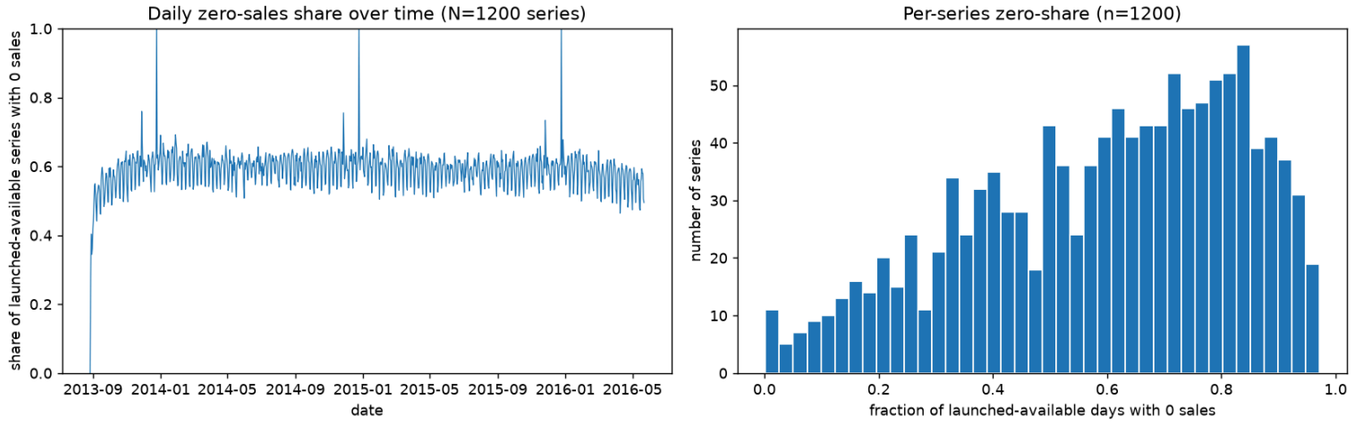
A clean but zero-heavy subset: most series are intermittent

1,200
series: CA_1, TX_1, WI_1 × 400 items

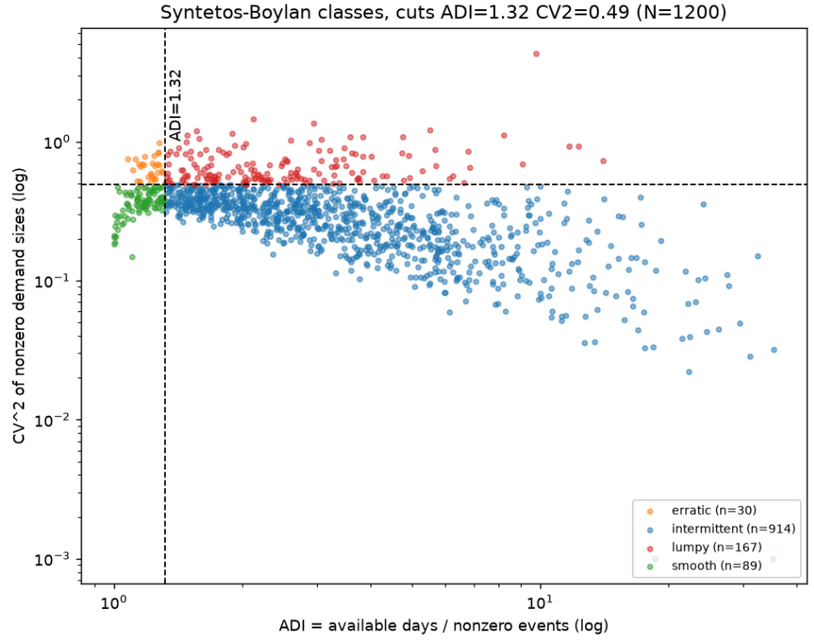
1,000
days, 2013-08-27 to 2016-05-22

95.64%
cells available (a price was posted)

0
duplicates, negatives or missing values



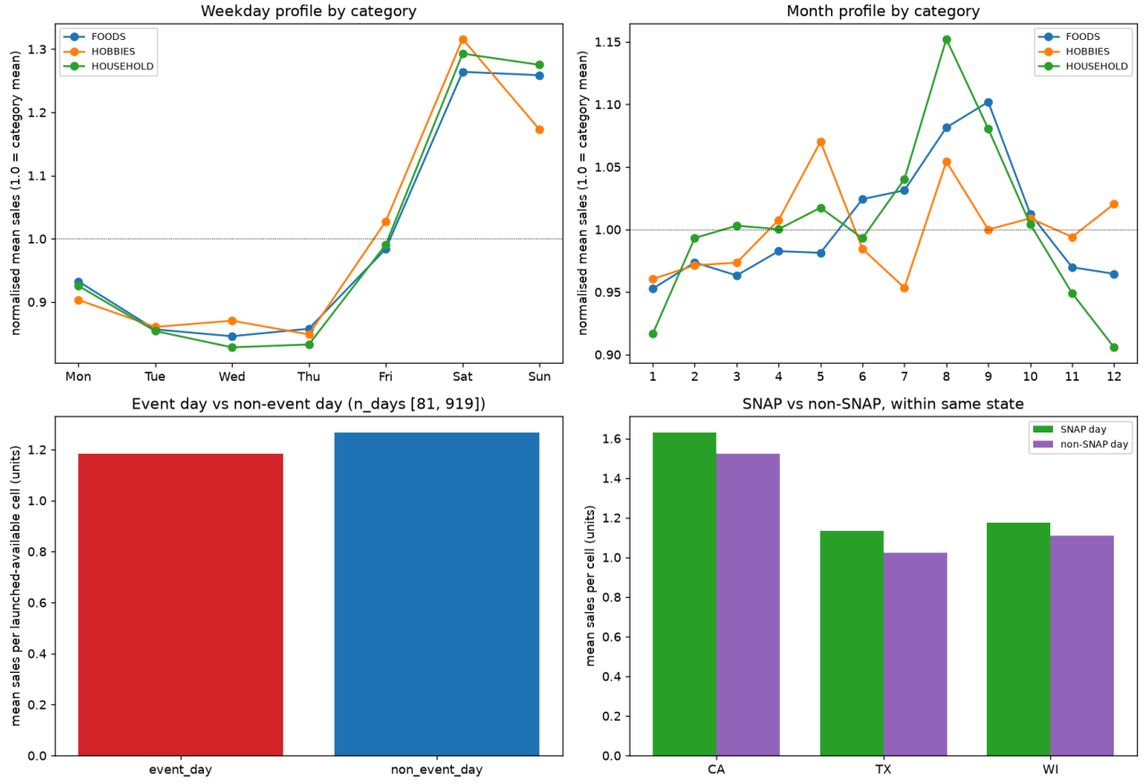
Daily zero-sales share (left) and per-series zero share (right). The low share at the start reflects series not yet launched.



Syntetos–Boylan classes: 914 intermittent, 167 lumpy, 89 smooth, 30 erratic.

EDA covers the full window, including test days. It is descriptive only and is not used as model input.

Weekdays drive seasonality; stockouts are inferred by a rule

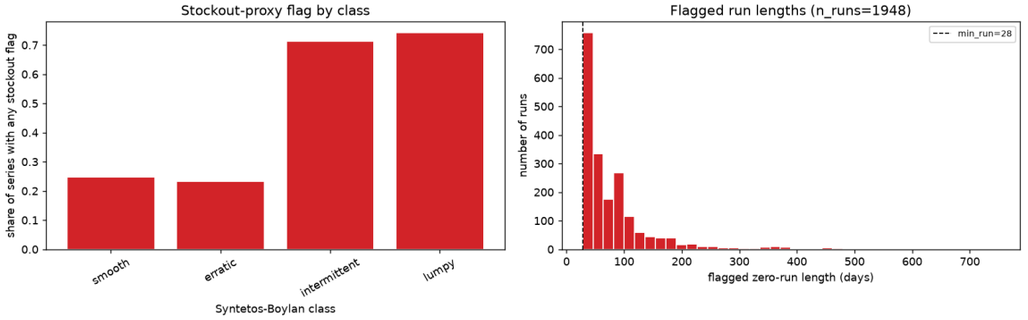


Weekend sales are 1.26–1.31× the category mean. Within each state, SNAP days are 1.058–1.107× non-SNAP days, and event days are 0.934× non-event days. All descriptive, not causal.

Stockout proxy (heuristic, not ground truth)

For a series with ≥ 10 sales, a zero run of length L is flagged if ***L ≥ max(28, 3 · ADI)***.

Flagged days are masked in training and **never imputed**. The rule can misfire both ways: it flags genuine quiet periods, and it misses short or partial outages.



The full window has 12.97% of cells flagged in 1,948 runs. Among scored test cells, 8.69% are flagged.

outputs/eda/weekday_profile.csv, snap_table.csv, event_table.csv, stockout_by_class.csv

FORECASTING APPROACHES

Five model families, one fit / predict interface

Kalman (local)

Per-series local level plus stochastic weekly seasonal, on √y. Noise ratios are picked by a per-series grid search over 16 pairs (not EM). Masked days skip the update step.

LightGBM (direct)

Four 7-day horizon buckets, each with one Poisson mean model and 7 quantile models: 32 boosters per origin, 200 trees each. No recursive feedback.

Kalman–LightGBM hybrid

LightGBM plus 3 Kalman features: filtered level, one-step sd, and the h-step state mean. It uses classical and ML parts only, with no deep learning.

DeepAR (global)

A shared 2-layer LSTM (64 units) with a negative binomial output and embeddings for item and hierarchy. The loss is masked on censored days. Quantiles come from 200 sampled paths.

Equal-weight ensemble

The ⅓ average of the Kalman, LightGBM and DeepAR quantiles, computed after forecasting with no extra training. Reference baseline: the 4-week seasonal naive.

Croston, SBA, TSB, ETS and ARIMA appear in the literature review only. They are not implemented baselines here.

History-only features plus covariates known in advance

59 LightGBM features (the hybrid adds 3)

GROUP	N	READ AT
Target lags y_t, y_{t-1}, y_{t-2}	3	anchor t
Same-weekday lags (4) and their mean	5	≤ t
Rolling mean, sd, nonzero rate, mask fraction over 7, 14, 28, 56, 112 days	20	window to t
Recency: days since sale, age, mean	3	anchor t
Calendar, events, SNAP	19	target day
Price features	4	target week
Static hierarchy, horizon h	5	–

Engineering safeguards

- The context object holds targets only up to the origin T_0.
- Masks are recomputed on that history slice.
- LightGBM training rows require $t + h \le T_0$.

Tested evidence

Scrambling all targets after the origin leaves the context and the forecasts of 6 model families unchanged (tolerance 10^{-5}). This rules out *target* leakage on the test fixture, not every possible leak.

Price assumption

Target-week prices assume M5 posts prices in advance. The whole-window mean used to fill unposted weeks is not strictly causal. Those cells are never scored.

Feature importance not produced: no trained booster was saved, and computing it needs a retrain.

EXPERIMENTAL SETUP

Four rolling origins, a 28-day horizon, three evaluation sets

Rolling origins

Test origins: 2016-01-31, 02-28, 03-27 and 04-24, 28 days apart, with H = 28 (the M5 horizon). One earlier validation origin, 2016-01-03, is reserved, but **tuning was not run**.

Evaluation sets (cell = origin × series × horizon)

All days, 1,116 established series	**124,992**
Non-censored: proxy-flagged cells removed	**114,127**
Simulated cold start: 84 series	**9,408**

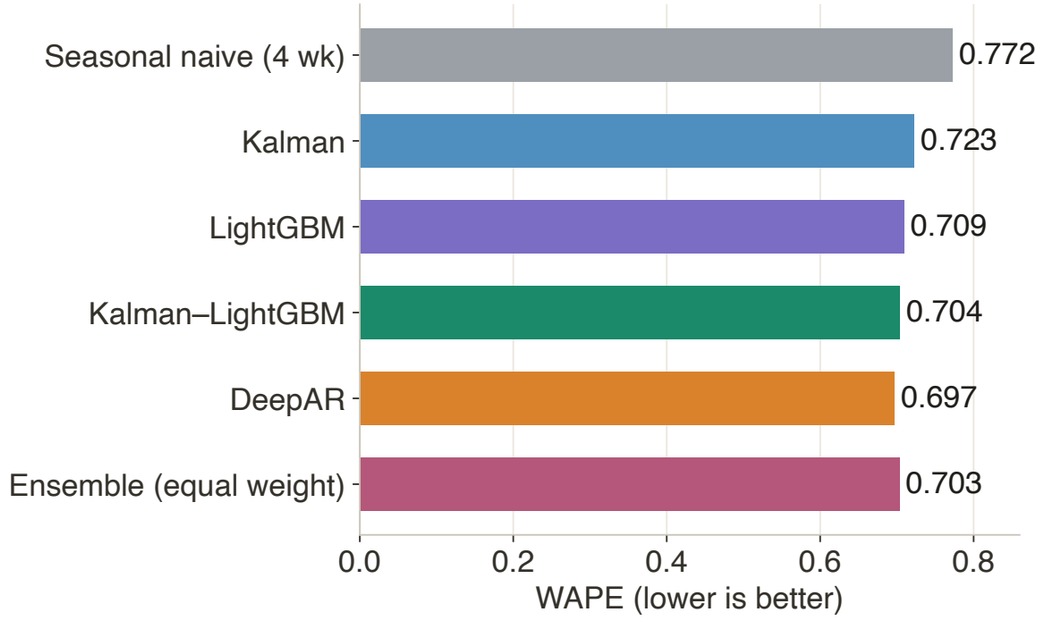
Cold start hides history from *existing* products: only the last 28 days stay visible and the item identity is unknown. It is not the same as a genuinely new launch.

Metrics (lower is better, except coverage)

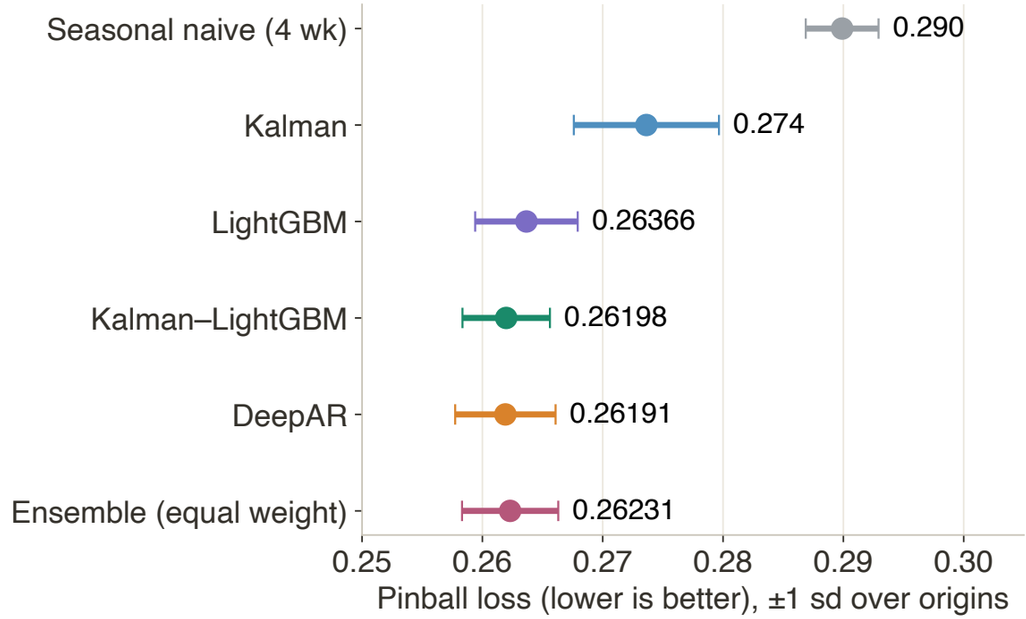
- **WAPE** = $\Sigma|y - \hat{y}_{0.5}| / \Sigma y$. A ratio of sums over all cells, using the median forecast.
- **Pinball** $\rho_q(u) = \max(qu, (q-1)u)$, averaged over 7 quantiles and all cells, unscaled.
- **MAE** uses the median forecast. **RMSE** uses the mean forecast.
- **Cov80 / Cov90**: share of actuals inside the central interval, inclusive bounds (nominal 0.80 / 0.90).
- **Significance**: paired bootstrap over series against the 4-week seasonal naive, 1,000 resamples.

Every model sees the same series, origins and masks, and is scored by the same evaluator.

DeepAR has the lowest WAPE; the top three are tied on pinball loss



WAPE over 124,992 cells. The axis starts at 0.

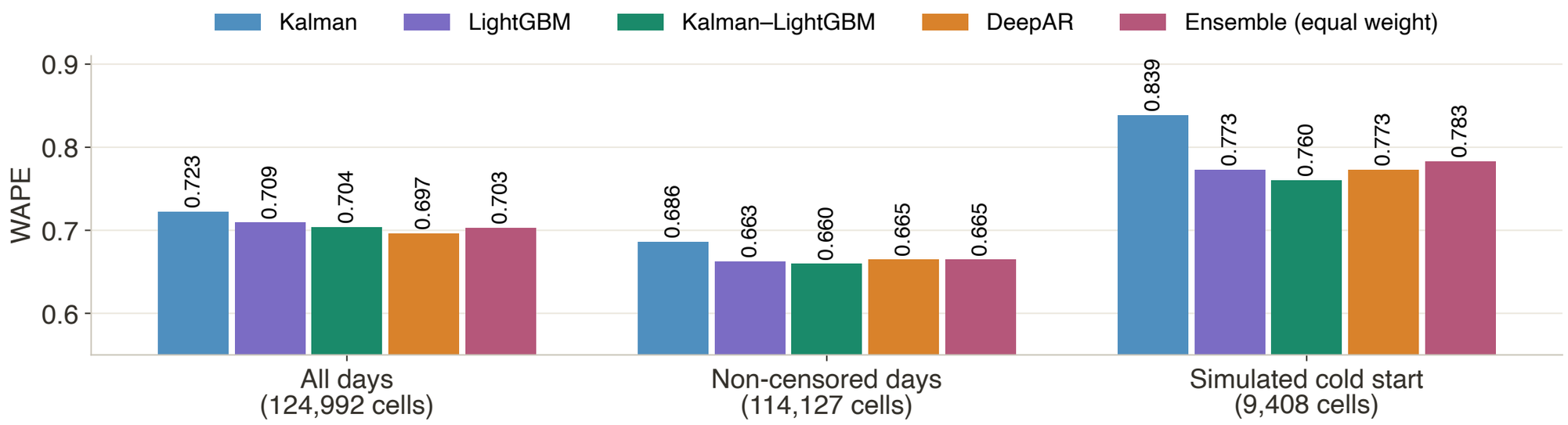


Pinball loss with ±1 std across the 4 origins. The gaps among the top three (≤ 0.0004) are about a tenth of that spread (≈ 0.004).

DeepAR: lowest WAPE (0.697) and MAE (0.891). Ensemble: lowest RMSE (1.782). All models beat the reference significantly (bootstrap CI excludes 0). No decisive winner on pinball.

RESULTS BY EVALUATION SETTING

The hybrid leads once flagged days are removed



WAPE, lower is better. The y-axis starts at 0.55 to make the gaps readable.

All days

DeepAR is best on WAPE (0.697). The hybrid is at 0.704 and the ensemble at 0.703.

Non-censored days

The hybrid is best on both metrics: WAPE 0.660, pinball 0.269. DeepAR drops to third.

Simulated cold start

The hybrid has the lowest WAPE (0.760), with LightGBM and DeepAR tied at 0.773. LightGBM has the lowest cold-start pinball (0.274).

main_all_days.csv · main_non_censored.csv · cold_start.csv. Kalman runs about 29× faster than LightGBM (13.7 s vs 397 s).

ABLATIONS AND CALIBRATION

NegBin beats Gaussian; masking helps on non-flagged days

Completed

7 Kalman and 2 DeepAR variants

VARIANT	WAPE	PINBALL	PINBALL, NON-CENS.
Kalman base (√, per-series grid)	0.723	0.2736	0.2822
Kalman, shared ratio	**0.718**	**0.2686**	**0.2772**
Kalman, no transform	0.786	0.2792	0.2857
Kalman, no masking	0.701	0.2698	0.2907
DeepAR base (NegBin, masked)	0.697	0.2619	0.2723
DeepAR, no loss masking	0.677	0.2565	0.2760
DeepAR, Gaussian likelihood	0.792	0.2784	0.2853

Fixed ratio, log1p and 180-day window rows: paper Table IV. DeepAR base = main run.

Reading the masking rows

Unmasked models learn the flagged zeros, so all-days scores improve but non-censored scores worsen. The reading assumes most flags are real stockouts.

Likelihood

A Gaussian instead of a negative binomial makes DeepAR's WAPE 12.1% worse and its pinball 5.9% worse.

Calibration caveat

DeepAR Cov90 = **0.965** (inclusive bounds; ties at zero inflate it). The discrete-aware check puts DeepAR in band at all 7 levels. Calibration is not solved.

outputs/ablation_kalman/ · outputs/ablation_deepar_quick/ · calibration.csv

LIMITATIONS AND THREATS TO VALIDITY

What these results do not show

Proxy labels

Stockout flags come from a heuristic rule. They shape both training and the non-censored evaluation set.

Narrow scope

One retailer, 3 stores, 1,200 series, 4 origins all in early 2016, a single seed (42).

No tuning

Hyperparameters are hand-set. Tuning could reorder models whose scores are this close.

Simulated cold start

Hiding history from existing items is not the same as a real product launch.

Covariate assumptions

Prices are assumed to be posted in advance, and the fill for weeks with no price is not strictly causal.

Missing baselines

Croston, SBA, TSB, ETS and ARIMA were not run.

Not run: the LightGBM ablation, the stockout-rule ablation, the other DeepAR variants, tuning, multiple seeds and feature importance.

Which model is best depends on the evaluation set

Conclusions

- Every learned model beats the 4-week seasonal naive (bootstrap CI excludes 0).
- DeepAR has the lowest all-days WAPE. DeepAR, the hybrid and the ensemble are tied on pinball.
- The hybrid leads on non-censored days and on simulated cold start.
- Kalman matches a 28-day moving average and runs about 29× faster than LightGBM. Its main value here is as features for the hybrid.

Future work

The remaining ablations, tuning on the validation origin, multiple seeds, feature importance, Croston/SBA/TSB baselines, and recalibrating DeepAR's quantiles.

Key references

1. Makridakis, Spiliotis, Assimakopoulos. M5 accuracy competition: Results, findings, and conclusions. *IJF* 38(4), 2022.
2. Salinas et al. DeepAR: Probabilistic forecasting with autoregressive recurrent networks. *IJF* 36(3), 2020.
3. Ke et al. LightGBM: A highly efficient gradient boosting decision tree. *NeurIPS* 30, 2017.
4. Durbin, Koopman. *Time Series Analysis by State Space Methods*, 2nd ed., OUP, 2012.
5. Syntetos, Boylan, Croston. On the categorization of demand patterns. *JORS* 56(5), 2005.
6. Gneiting, Raftery. Strictly proper scoring rules, prediction, and estimation. *JASA* 102(477), 2007.
7. Montero-Manso, Hyndman. Forecasting groups of time series: locality and globality. *IJF* 37(4), 2021.

The full list of 19 verified references is in the paper, with DOIs.